PROGRAMMING 2B (PROG6212)

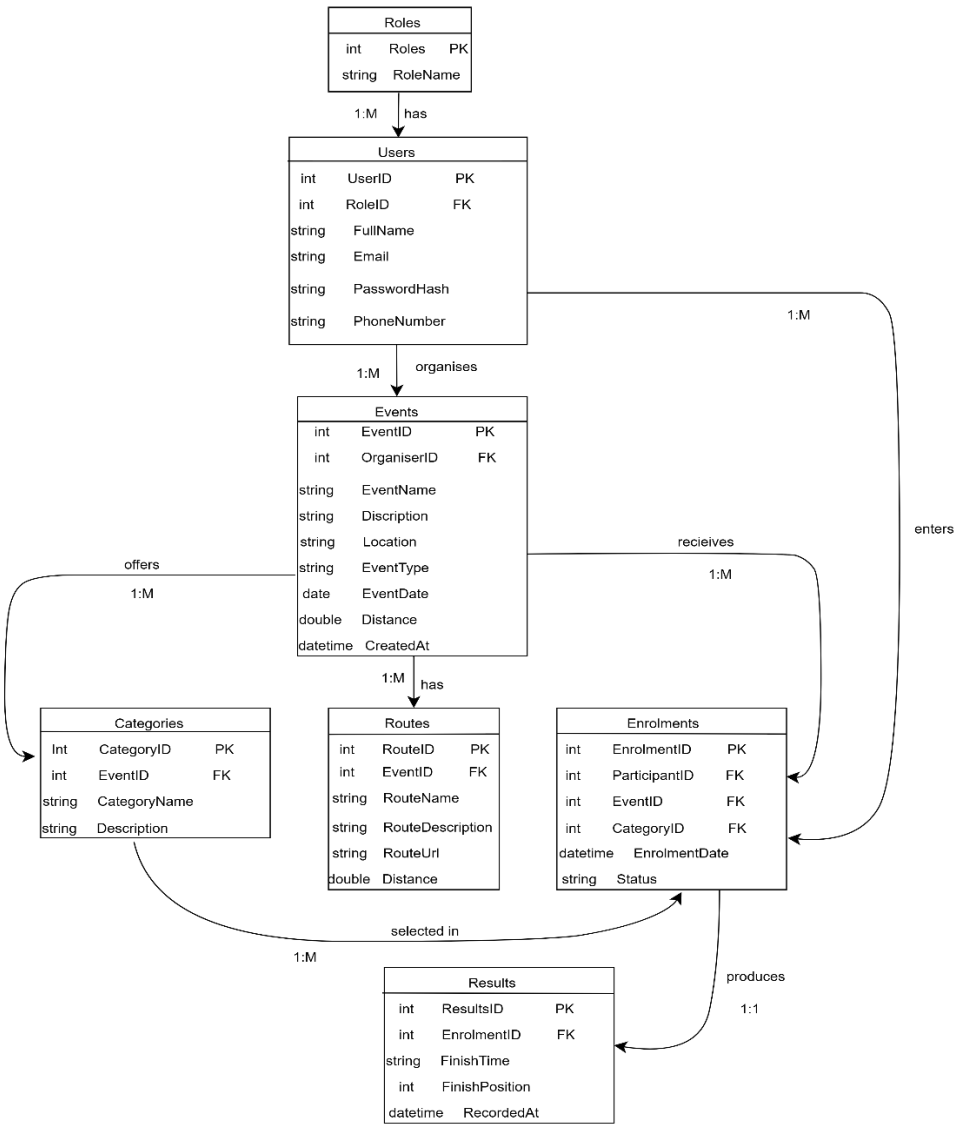
POE – part 1

System Planning and Database

By Sasha Tizora (ST10497496)

SECTION A

There are six entities on my ERD: Roles, Users, Events, Categories, Enrolments, and Results. There are also attributes, PKs, FKs, relations, and cardinality of each entity. The diagram depicts the way the Organiser creates an event, the Participant enrolls in an event and a category, and the Result is made for the completed enrollment.



SECTION B: API Endpoint Plan

The following table illustrates the API endpoints that have been designed for the RaceDay project. The API table shows the method to use, the route, the purpose of each endpoint, the user role required, the request body, and the expected response for each endpoint.

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
POST	/api/auth/register	Creates a new account as either an Organiser or Participant	None	{ fullName, email, password, phoneNumber, role }	201 Created – user created; 400 Bad Request – invalid data; 409 Conflict – email already registered
POST	/api/auth/login	Authenticates a user and starts a session with their role	Nonee	{ email, password }	200 OK – session started, returns role; 401 Unauthorized – invalid credentials
POST	/api/auth/logout	Ends the current user's session	Any	none	200 OK – session ended
GET	/api/profile	Returns the logged-in user's own profile information	Any	none	200 OK – profile data; 401 Unauthorized

PUT	/api/profile	Updates the logged-in user's own profile information	Any	{ fullName, phoneNumber }	200 OK – updated profile; 400 Bad Request; 401 Unauthorized
GET	/api/events	Returns a list of all upcoming RaceDay events	None	none	200 OK – list of events
GET	/api/events/{id}	Returns full details for a specific event using its event ID	None	none	200 OK – event details; 404 Not Found
POST	/api/events	Creates a new RaceDay event	Organiser	{ eventName, description, eventDate, location, distance, eventType }	201 Created – event created; 400 Bad Request; 401/403 if access invalid
PUT	/api/events/{id}	Updates an existing event owned by the Organiser	Organiser	{ eventName, description, eventDate, location, distance, eventType }	200 OK – updated event; 403 Forbidden – not the owner; 404 Not Found
DELETE	/api/ events/{id}	Deletes an event owned by the Organiser	Organiser	none	200 OK – deleted; 403 Forbidden; 404 Not Found

GET	/api/events/{id}/categories	Returns all categories available for a specific event	None	none	200 OK – list of categories; 404 Not Found
POST	/api/events/{id}/categories	Adds a new age or distance category to an event	Organiser	{ categoryName, description }	201 Created – category created; 403 Forbidden; 404 Not Found
PUT	/api/categories/{id}	Updates an existing category	Organiser	{ categoryName, description }	200 OK – updated; 403 Forbidden; 404 Not Found
DELETE	/api/categories/{id}	Deletes a category from an event	Organiser	none	200 OK – deleted; 403 Forbidden; 404 Not Found
GET	/api/events/{id}/routes	Returns all route details for a specific event	None	none	200 OK – list of routes; 404 Not Found
POST	/api/events/{id}/routes	Adds a new route (course details, GPX link, elevation) to an event	Organiser	{ routeName, routeDescription, routeUrl, distance }	201 Created – route added; 403 Forbidden; 404 Not Found

PUT	/api/routes/{id}	Updates an existing route for an event	Organiser	{ routeName, routeDescription, routeUrl, distance }	200 OK – updated; 403 Forbidden; 404 Not Found
DELETE	/api/routes/{id}	Deletes a route from an event	Organiser	none	200 OK – deleted; 403 Forbidden; 404 Not Found
POST	/api/enrolments	Enrols the logged-in Participant into an event under a chosen category	Participant	{ eventId, categoryId }	201 Created – enrolment recorded; 400 Bad Request; 409 Conflict – already enrolled
GET	/api/enrolments/my	Returns all enrolments belonging to the logged-in Participant	Participant	None	200 OK – list of enrolments
GET	/api/events/{id}/enrolment	Retrieves all Participants enrolled in an Organiser's event.	Organiser	None	200 OK – list of enrolments; 403 Forbidden
POST	/api/results	Records a Participant's finish time and finishing position.	Organiser	{enrolmentId, finishTime, finishPosition}	201 Created – result recorded; 403 Forbidd

					en; 404 Not Found
GET	/api/results/my	Retrieves the logged-in Participant's race results.	participant	None	200 OK – list of results